

Day 14 : Shared Preference and Local Storage

1. Introduction to Shared Preferences

SharedPreferences is a Flutter package used to store small amounts of data permanently on a device using key-value pairs.

Think of it like a small notebook where:

name = Sneha

age = 20

isLogin = true

theme = dark

Even if the app closes, this data remains saved.

2. Why Do We Use it?

Without SharedPreferences:

- User logs in → App closes → Login lost
- User selects Dark Theme → App closes → Theme resets

With SharedPreferences:

- Login state remains saved
- Theme remains saved
- Settings remain saved
- Language preference remains saved

3. Key-Value Storage

SharedPreferences stores data as:

Key	Value

name	Sneha
age	20
darkMode	true
cgpa	8.5

1. Key

A unique name used to identify data.

'name'

'age'

'darkMode'

2. Value

Actual stored data.

'Sneha'

20

true

EXAMPLE:

```
prefs.setString('name', 'Sneha');
```

4. Setup SharedPreferences?

Step 1: Add Dependency

In pubspec.yaml

```
dependencies:  
  flutter:  
    sdk: flutter  
  shared_preferences: ^2.5.3
```

Step 2: Install Package

```
flutter pub get
```

Step 3: Import Package

```
import 'package:shared_preferences/shared_preferences.dart';
```

5. Creating SharedPreferences Object

Before storing or retrieving data:

```
SharedPreferences prefs =  
  await SharedPreferences.getInstance();
```

6. Store

1. Store String

```
prefs.setString(  
    'name',  
    'Sneha'  
);
```

2. Store Integer

```
prefs.setInt(  
    'age',  
    20  
);
```

3. Store Double

```
prefs.setDouble(  
    'cgpa',  
    8.6  
);
```

4. Store List

```
prefs.setStringList(  
    'subjects',  
    ['MAD', 'NIS', 'Python']  
);
```

7. Retrieve Data

Retrieving means reading saved data.

1. Get String

```
String? name =  
    prefs.getString('name');
```

2. Get Int

```
int? age =  
    prefs.getInt('age');
```

3. Get Bool

```
bool? login =  
    prefs.getBool('isLogin');
```

4. Get Double

```
double? cgpa =  
    prefs.getDouble('cgpa');
```

5. Get List

```
List<String>? subjects =  
    prefs.getStringList('subjects');
```

8. Null Values

Suppose key does not exist:

```
String? name =  
    prefs.getString('name');
```

1. Default Value

```
String name =  
    prefs.getString('name') ?? 'Guest';
```

9. Remove Data

```
prefs.remove('name');
```

10. Login State Persistence

Save Login

```
prefs.setBool(  
    'login',  
    true  
);
```

Check Login

```
bool login =  
    prefs.getBool('login') ?? false;
```

11. Theme Persistence

```
prefs.setBool(  
    'darkMode',  
    true  
);
```

12. Language Preference

```
prefs.setString(  
    'language',  
    'English'  
);
```

Retrive

```
String language =  
prefs.getString('language')  
?? 'English';
```

Output :



